

Medical Image Classification Using SVM, CNN, and XGBoost: A Comparative Study on Brain MRI and Chest X-ray Datasets - TANISHK G

Abstract: Medical image classification has become an important application of artificial intelligence in healthcare, enabling automated analysis of medical scans and supporting clinical decision-making. This project investigates the effectiveness of multiple machine learning, deep learning, and hybrid approaches for the classification of Brain MRI (multi-class) and Chest X-Ray (binary-class) images. The evaluated methods include Support Vector Machine (SVM) with Principal Component Analysis (PCA), Convolutional Neural Networks (CNN) using transfer learning, XGBoost, PCA + XGBoost, and VGG16 + XGBoost. Traditional machine learning methods were assessed alongside deep learning models to examine their suitability for medical image analysis. Additionally, hybrid approaches combining feature extraction and ensemble learning techniques were investigated to determine their effectiveness in capturing complex image characteristics.

The results demonstrate that increasing the amount of training data generally improves classification performance and model generalization. While traditional machine learning methods provide reliable classification capability, deep learning approaches are more effective at learning discriminative image features directly from medical scans. Furthermore, hybrid models that combine deep feature extraction with ensemble learning achieve strong and consistent performance across both datasets.

Overall, the findings highlight the importance of feature representation in medical image classification and demonstrate the potential of deep learning and hybrid techniques for automated disease detection. The study provides a comprehensive comparison of multiple classification strategies and offers insights into their applicability for medical imaging tasks.

1. INTRODUCTION

Medical imaging is a vital component of modern healthcare that enables the visualization of internal body structures without invasive procedures. It plays an important role in disease diagnosis, treatment planning, and patient monitoring by providing detailed information about organs, tissues, and physiological processes. Medical imaging allows healthcare professionals to detect abnormalities at an early stage, improving diagnostic accuracy and patient outcomes.

Several imaging modalities are commonly used in clinical practice, including X-Ray, Computed Tomography (CT), Ultrasound, Positron Emission Tomography (PET), and Magnetic Resonance Imaging (MRI). Each modality serves a specific purpose depending on the medical condition being investigated. MRI is widely used for brain imaging due to its excellent soft tissue contrast, while Chest X-Ray imaging remains one of the most frequently used diagnostic tools for identifying respiratory and thoracic diseases.

The increasing volume of medical imaging data generated in healthcare institutions has further emphasized the need for automated image analysis techniques.

Artificial Intelligence (AI) has emerged as a powerful tool for medical image analysis, enabling computers to identify patterns and abnormalities within medical scans. AI-based systems can assist healthcare professionals by improving diagnostic efficiency, reducing workload, and supporting clinical decision-making.

Machine Learning (ML) algorithms such as Support Vector Machines (SVM) and XGBoost utilize numerical image features to perform classification tasks. In contrast, Deep Learning (DL) models, particularly Convolutional Neural Networks (CNNs), automatically learn relevant image features directly from the data. This capability has made deep learning one of the most successful approaches for medical image classification.

Recent research has also explored hybrid approaches that combine deep feature extraction with traditional machine learning classifiers. Such methods aim to leverage the strengths of both paradigms and achieve improved classification performance. In this project, traditional machine learning, deep learning, and

hybrid techniques are evaluated on Brain MRI and Chest X-Ray datasets to investigate their effectiveness for automated medical image classification.

2. Dataset:

2.1 Brain MRI dataset: The Brain MRI dataset used in this study consists of brain scan images belonging to four classes: Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and No Tumor. The dataset is designed for multi-class classification, where the objective is to correctly identify the category of a given MRI image.

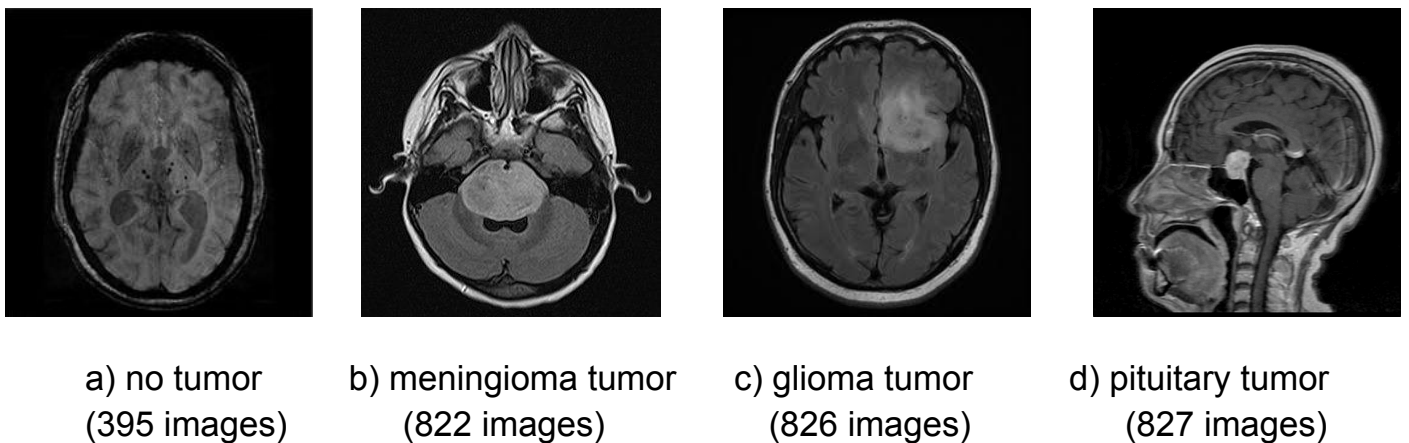


Figure 1. Four distinct classes of brain tumor dataset

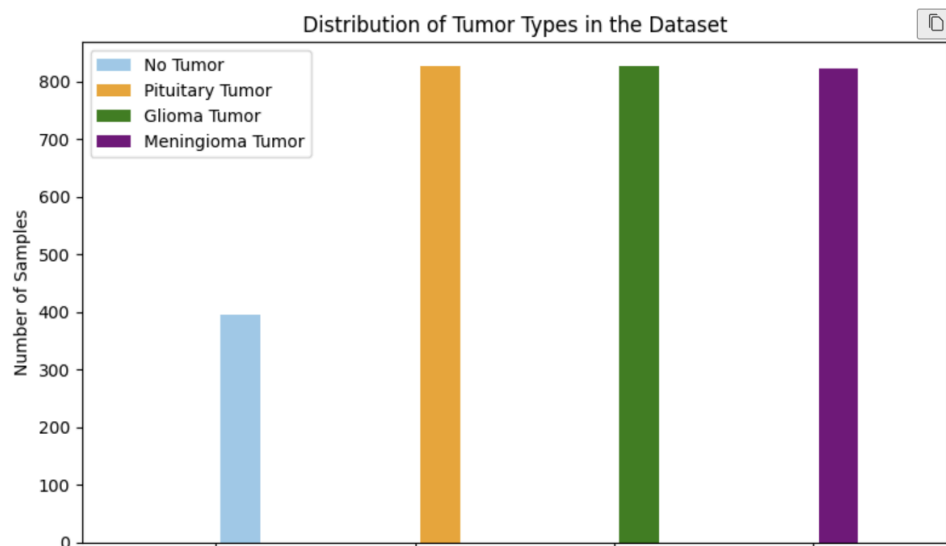
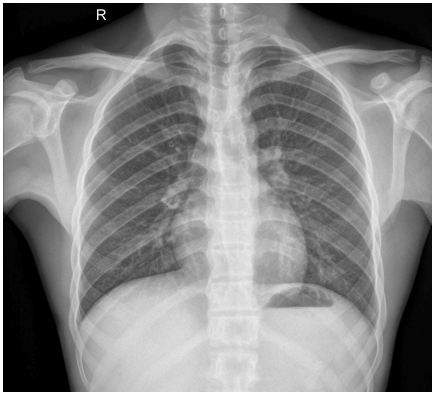
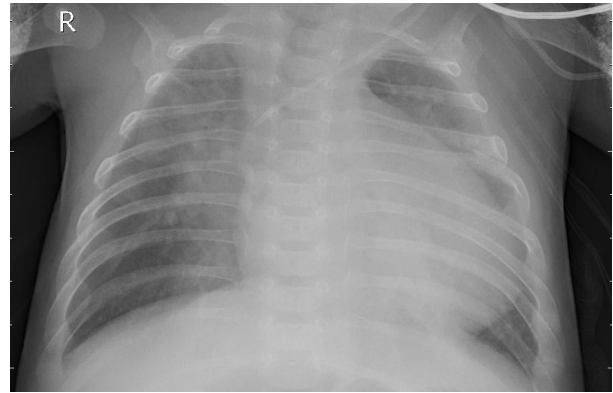


Figure 2: Histogram representation of Brain MRI data

2.2 Chest X-ray dataset: The Chest X-ray dataset used in this study consists of chest radiographic images belonging to two classes: Normal and Pneumonia. The dataset is designed for binary classification, where the objective is to correctly identify whether a given chest X-ray image indicates a healthy subject or a subject affected by pneumonia.



a) Normal (1583 images)



b) Pneumonia (4273 images)

Figure 3. Two distinct class of chest X-ray dataset

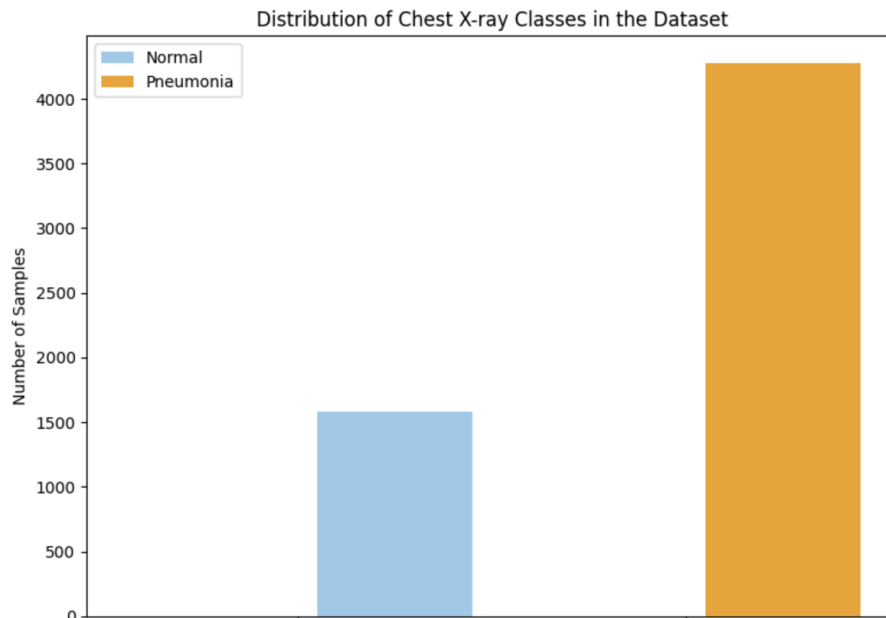


Figure 4: Histogram representation of Chest X-Ray data

3. Evaluation metrics

3.1 Confusion Matrix: A confusion matrix is a table that summarizes the model's predictions by comparing predicted labels with actual labels. Size $n \times n$ where n is the number of classes.

- True Positive (TP) : Positive class correctly predicted as positive
- False positive (FP) : Positive class incorrectly predicted as positive
- False Negative (FN) : Negative class incorrectly predicted as negative
- True Negative (TN) : Negative class correctly predicted as negative

		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

Figure 5: Confusion Matrix

Based on the values of TP, FP, FN and TN in the confusion matrix other evaluation metrics such as accuracy, precision, recall (sensitivity), specificity and F1-score can be calculated.

3.2 Accuracy: Measures the proportion of data samples for which the model's prediction matches the actual class label.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

3.3 Precision: Measures the proportion of positive predictions made by the model that are actually correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

3.4 Recall or Sensitivity(True positive rate): Measures the proportion of positive data samples that are correctly identified by the model.

$$\text{Recall} = \text{TP} / \text{TP} + \text{FN}$$

3.5 Specificity (True negative rate): Measures the proportion of negative data samples that are correctly identified by the model.

$$\text{Specificity} = \text{TN} / \text{TN} + \text{FP}$$

3.6 F1-score: Measures how well the model balances correct positive predictions and the identification of positive samples.

$$\text{F1-score} = 2 * ((\text{recall} * \text{precision}) / (\text{recall} + \text{precision}))$$

3.7 Receiver operating characteristic (ROC) curve: It is a graphical plot that illustrates the performance of a classification model by plotting the True Positive Rate against the False Positive Rate at various threshold settings.

The Area Under the ROC Curve (AUC) measures the model's overall ability to distinguish between classes. Higher AUC value indicates better discriminative performance.

3.8 Validation loss: Validation loss measures the error of the model on unseen validation data during training. Lower validation loss indicates better generalization and predictive performance.

3.9 Loss Curve: The loss curve is a graphical representation of training and validation loss across epochs. It is used to monitor the learning process and identify issues such as underfitting or overfitting.

4. METHODOLOGY

4.1 SUPPORT VECTOR MACHINE (SVM) : Support Vector Machines (SVM) is a supervised machine learning algorithm widely used for classification

tasks. The primary objective of SVM is to find an optimal hyperplane that maximizes the margin between different classes, by doing so SVM aims to improve the model's ability to generalize unseen data.

Unlike deep learning models, SVM cannot directly understand image structure and therefore requires images to be converted into numerical feature vectors. Since each image contains thousands of pixel values, **Principal Component Analysis (PCA)** was used to reduce dimensionality before classification.

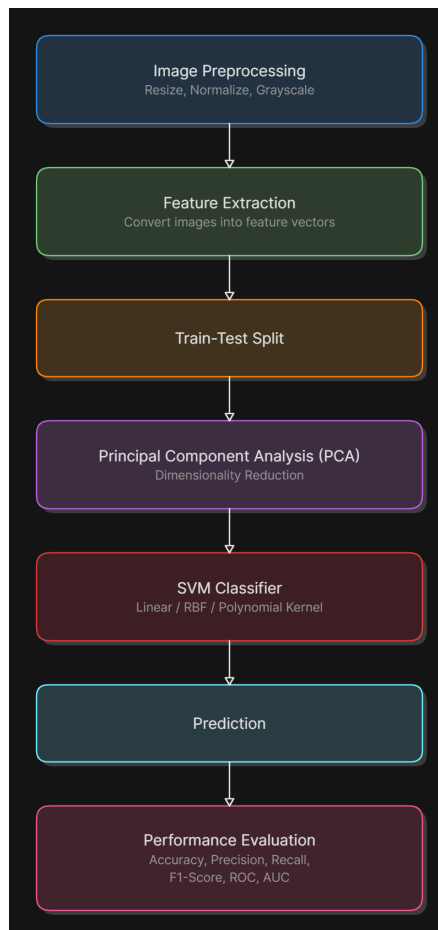


Figure 6: SVM + PCA pipeline flowchart

4.1.1 RESULTS

	Train/Test Split	Accuracy		Recall		Specificity		F1 Score		Precision	
		RBF	Lin	RBF	Lin	RBF	Lin	RBF	Lin	RBF	Lin
brain	20/80 (574/2296)	72.82	72.34	72.82	72.34	90.72	90.54	71.53	71.95	72	72
	40/60 (1148/1722)	78.69	76.77	78.69	76.77	92.76	92.1	77.96	76.38	79	76
	60/40 (1722/1148)	81.71	80.84	81.71	80.84	93.8	93.5	81.21	80.53	82	81
	80/20 (2296/574)	83.1	81.36	83.1	81.36	94.31	93.7	82.7	81.08	83	81
chest	20/80 (1171/4685)	94.66	92.44	94.66	92.44	92.31	89.98	94.61	92.41	95	92
	40/60 (2342/3514)	95.11	93.06	95.11	93.06	93	90.69	95.06	93.01	95	93
	60/40 (3514/2342)	95.73	93.3	95.73	93.3	93.92	91.03	95.7	93.26	96	93
	80/20 (4685/1171)	96.16	93.52	96.16	93.52	94.65	91.48	96.14	93.49	96	93

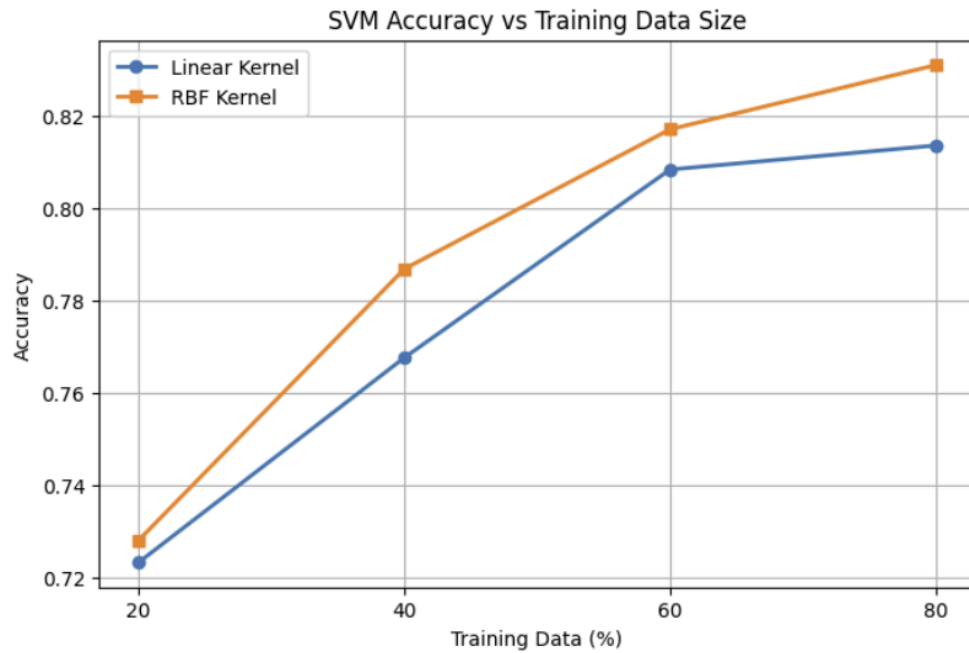


Figure 7: Linear vs RBF kernel (accuracy comparison against training data)

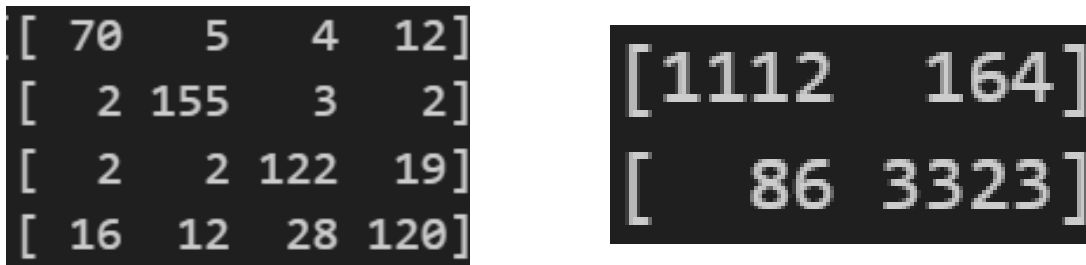


Figure 8: Confusion matrix for SVM RBF kernel (20% training) a) brain b) chest

4.1.2 INFERENCE

1. Classification accuracy increased consistently as the proportion of training data increased for both Brain MRI and Chest X-Ray datasets, indicating that the models benefited from additional training samples and improved generalization.
2. Classification accuracy increased consistently as the proportion of training data increased for both Brain MRI and Chest X-Ray datasets, indicating that the models benefited from additional training samples and improved generalization.
3. The Chest X-Ray dataset achieved higher classification accuracy compared to the Brain MRI dataset. This indicates that the extracted features in the Chest X-Ray images were more distinguishable, making the classification task comparatively easier.
4. The 80:20 train-test split produced the highest classification performance for both kernels, demonstrating that larger training sets enable the model to learn more representative decision boundaries while still retaining sufficient test samples for evaluation.

4.2 Convolutional Neural Network (CNN) : Convolutional Neural Networks (CNNs) are a class of deep learning models specifically designed for image analysis tasks. Unlike traditional machine learning algorithms that rely on

manually extracted features, CNNs automatically learn important visual patterns directly from the input images through a series of convolution and pooling operations. This ability to learn hierarchical features makes CNNs highly effective for image classification problems.

CNNs were applied to both the Brain MRI and Chest X-Ray datasets to classify medical images based on the presence of different pathological conditions. By learning features directly from the images, the CNN eliminates the need for manual feature extraction techniques. For Brain MRI images, the model learns characteristics associated with different tumor types, while for Chest X-Ray images it identifies visual patterns indicative of normal and abnormal lung conditions.

A pre-trained MobileNet architecture was employed using transfer learning for image classification. Transfer learning allows the model to leverage previously learned features and adapt them to the Brain MRI and Chest X-Ray classification tasks. MobileNet was used as a feature extractor, enabling the model to learn meaningful representations and identify disease-specific patterns.

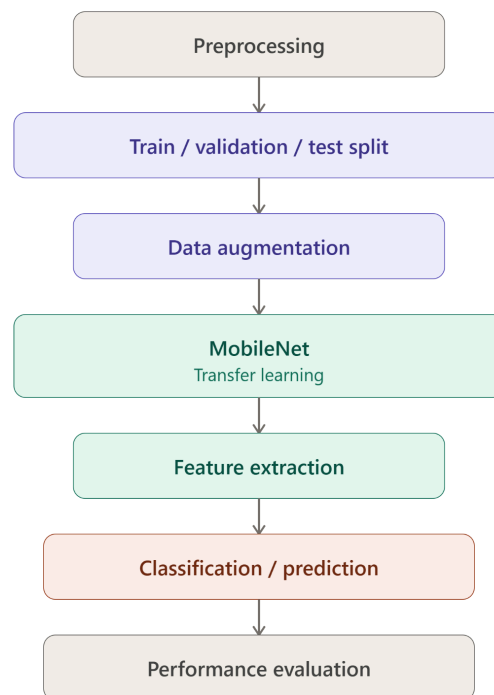


Figure 9: CNN pipeline flowchart

4.2.1 RESULTS

brain	Train/val/test split	Lowest val loss (epoch)	Best val accuracy (%)	Test accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	AUC
	20/5/15 (574/145/431)	0.266 (47)	91.03 (49)	89.1	89	89	89	—
	40/5/15 (1146/145/431)	0.3232 (55)	88.97 (46)	87.7	88	88	88	—
	60/5/15 (1179/145/431)	0.2156 (54)	91.72 (60)	90.26	91	90	90	—
	80/5/15 (2294/145/431)	0.192 (39)	93.79 (38)	92.58	93	93	92	—
chest	Train/val/test split	Lowest val loss (epoch)	Best val accuracy (%)	Test accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	AUC
	20/5/15 (1171/293/878)	0.125 (51)	95.56 (51)	63.44	99.38	50.23	66.74	0.9106
	40/5/15 (2342/293/878)	0.1008 (57)	96.59 (57)	71.87	99.25	61.93	76.27	0.9548
	60/5/15 (3513/293/878)	0.0926 (43)	97.61 (43)	86.79	99.44	82.37	90.1	0.9845
	80/5/15 (4684/293/878)	0.0905 (46)	97.61 (42)	93.96	95.79	95.94	95.87	0.987

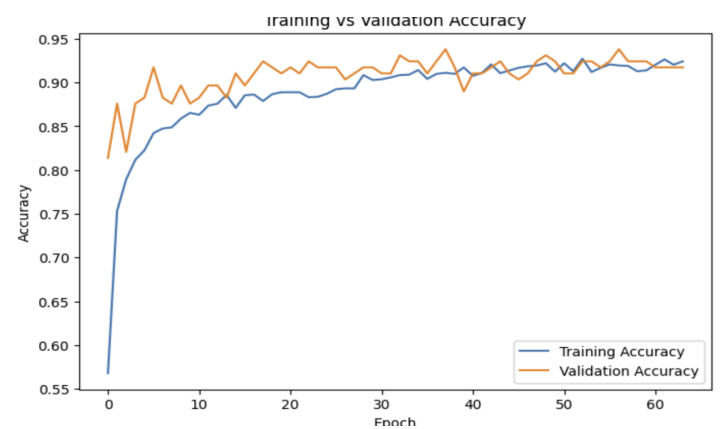
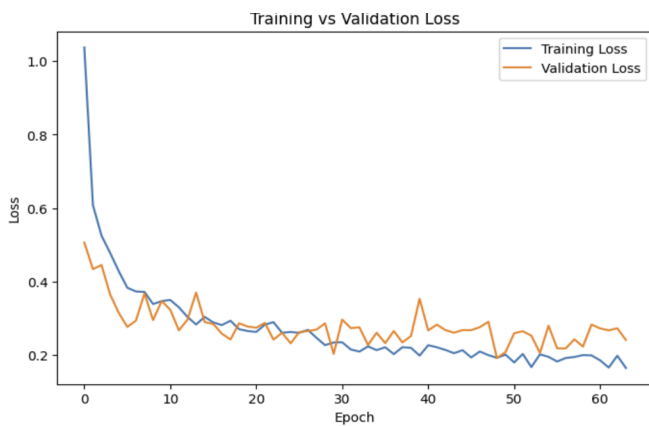


Figure 10: a) validation loss curve b) validation accuracy curve for multiclass (Brain)
(80/5/15 split)

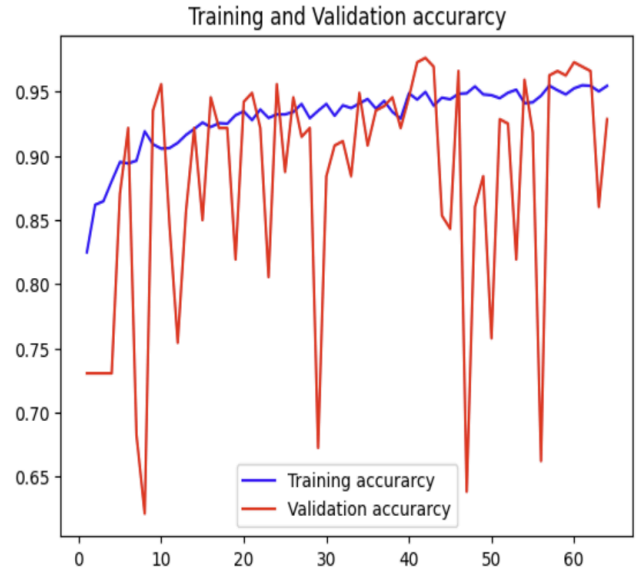
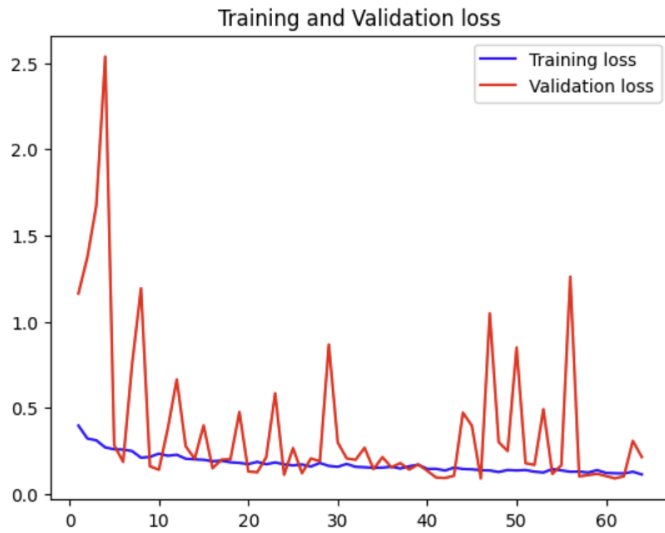


Figure 11: a) validation loss curve b) validation accuracy curve for binary class (chest)
(80/5/15 split)

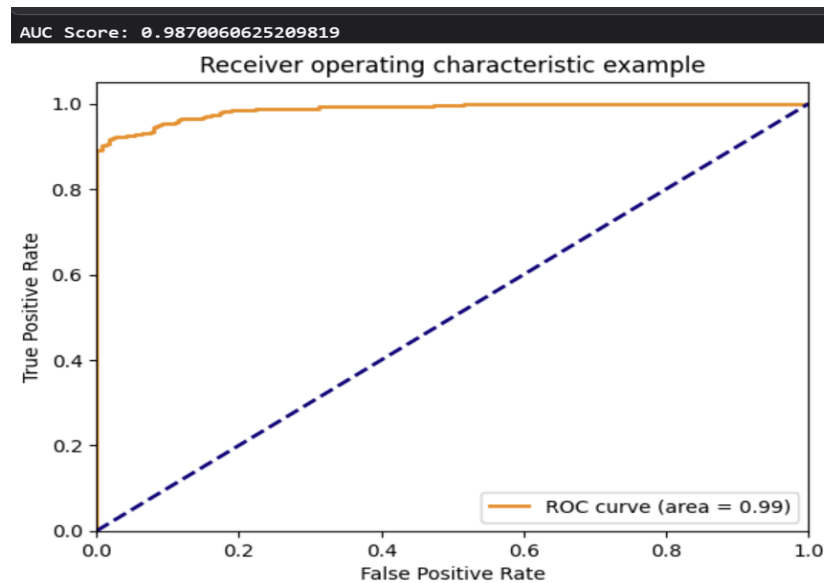


Figure 12: ROC curve - chest (80/5/15 split)

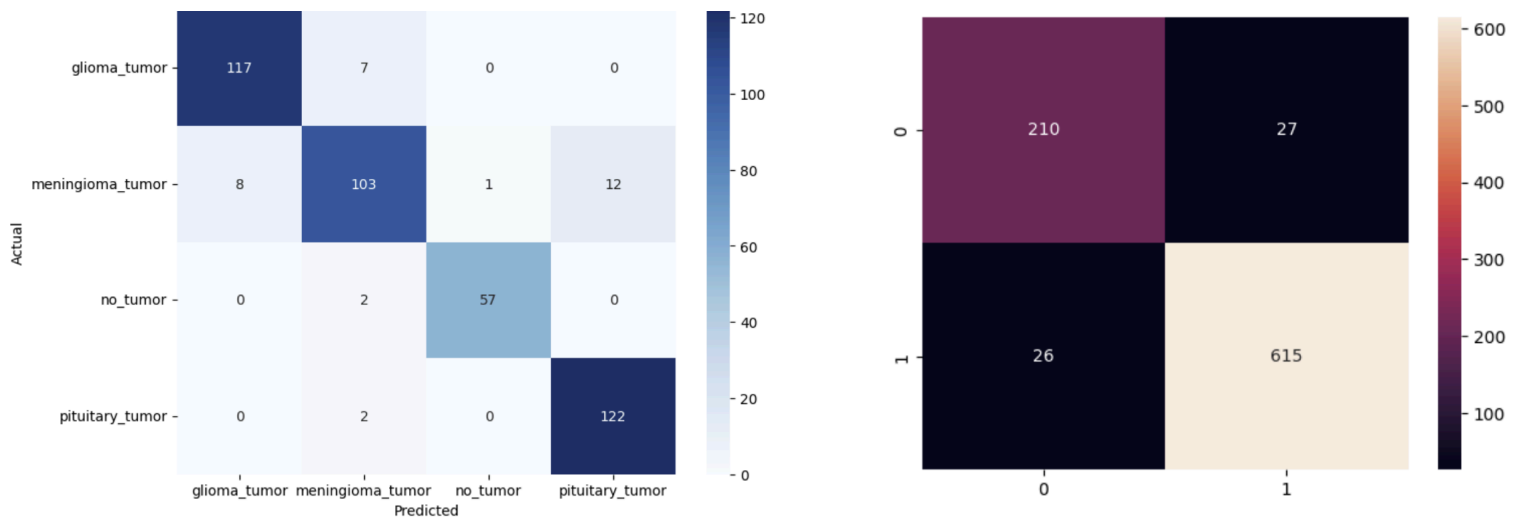


Figure 13: Confusion matrix of brain and chest for 80/5/15 split

4.2.2 INFERENCE:

1. Overall performance (for both brain MRI and chest X-Ray datasets) increased as we increased the training data indicating that the model was able to learn more representative image features.
2. The 80/5/15 split achieved the best overall performance for both datasets, producing the highest test accuracy and the most favorable validation metrics.
3. The epochs corresponding to the highest validation accuracy and lowest validation loss were either identical or very close, indicating stable convergence during training.
4. Validation loss generally decreased while validation accuracy increased with larger training splits, suggesting improved generalization and minimal overfitting.
5. The Chest X-Ray dataset achieved slightly higher performance than the Brain MRI dataset (93.96% vs 92.58% test accuracy), suggesting that

disease-specific patterns in Chest X-Ray (binary class) images were more easily distinguishable by the model.

6. The validation accuracy curve for the chest X-ray shows fluctuations which maybe due to the small validation dataset and use of data augmentation during training. The absence of a widening gap between training and validation accuracy suggests that the model maintained good generalization capability throughout the training process.

4.3 XGBoost: Extreme Gradient Boosting (XGBoost) is an advanced ensemble machine learning algorithm based on gradient-boosted decision trees. Instead of relying on a single classifier, XGBoost builds a sequence of decision trees where each new tree attempts to correct the errors made by the previous ones. This iterative learning process enables the model to capture complex patterns within the data while maintaining high predictive performance. Due to its efficiency, scalability, and ability to handle high-dimensional datasets, XGBoost has become one of the most widely used machine learning algorithms for classification and prediction tasks.

XGBoost was applied to both the Brain MRI and Chest X-Ray datasets for medical image classification. Since XGBoost cannot process raw images directly, image features were first extracted and converted into numerical feature vectors before training the classifier. The algorithm then constructed an ensemble of decision trees using a gradient boosting framework, where each successive tree aimed to reduce the errors of the previous trees. This enabled the model to learn complex relationships within the extracted image features and effectively distinguish between different medical image classes.

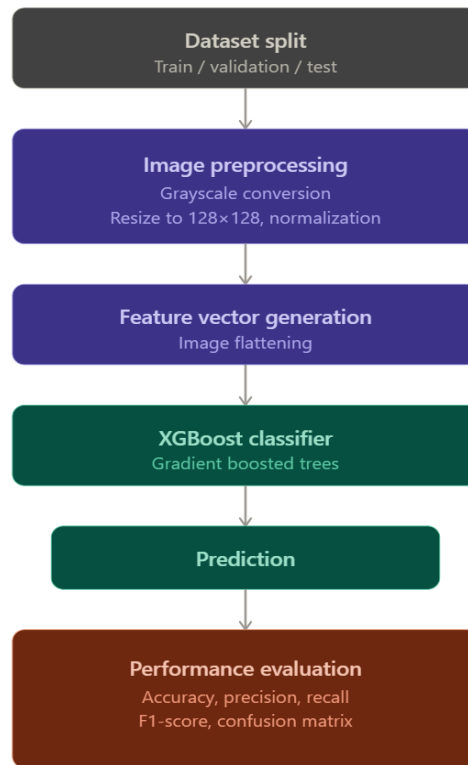


Figure 14: XGBoost pipeline flowchart

4.3.1 RESULTS:

brain	Train/val/test split	Validation accuracy	Test accuracy	Macro precision	Macro recall	Macro F1-score	Best boosting round
	20/5/15 (574/145/431)	81.38%	86.08%	0.86	0.86	0.86	88
	40/5/15 (1146/145/431)	87.59%	89.10%	0.88	0.89	0.88	168
	60/5/15 (1179/145/431)	91.72%	91.42%	0.91	0.91	0.91	165
	80/5/15 (2294/145/431)	91.72%	93.97%	0.94	0.94	0.94	181
chest	Train/val/test split	Validation accuracy	Test accuracy	Precision	Sensitivity (recall)	F1-score	Best boosting round
	20/5/15 (1171/293/878)	95.22%	92.14%	0.93	0.97	0.95	141
	40/5/15 (2342/293/878)	95.56%	93.74%	0.95	0.97	0.96	191
	60/5/15 (3513/293/878)	96.25%	94.99%	0.96	0.98	0.97	176
	80/5/15 (4684/293/878)	95.90%	94.76%	0.96	0.97	0.96	193

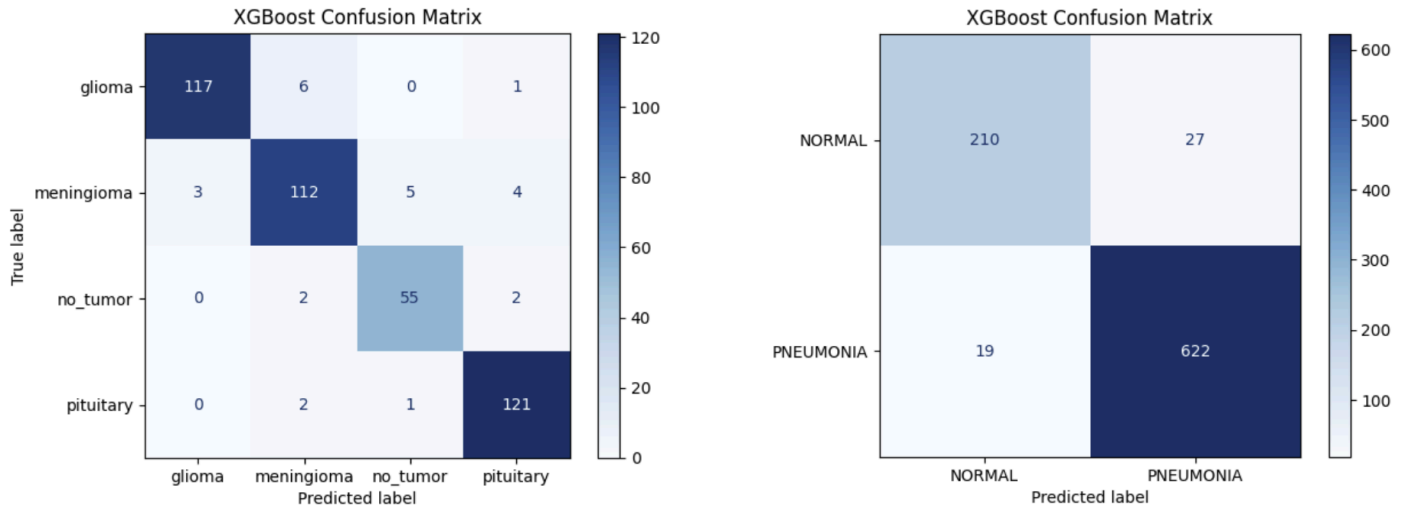


Figure 15: Confusion matrix of Brain MRI and Chest X-Ray for 80/5/15 split

4.3.2 INFERENCE

1. As expected, for both Brain MRI and Chest X-Ray datasets, classification performance improved as the amount of training data increased, indicating that XGBoost benefited from larger training sets and learned more representative feature patterns.
2. The consistently high precision and recall values across both datasets indicate that XGBoost was effective in correctly identifying disease classes while maintaining a low rate of incorrect predictions.
3. The increase in the best boosting rounds as training data increased shows that the model was able to utilize additional boosting iterations to learn more complex feature relationships, contributing to improved classification performance.
4. In the chest X-Ray dataset performance improved significantly from 20% to 60% training data, achieving the highest test accuracy of 94.99% and F1-score of 0.97 at the 60/5/15 split. The slight decrease at the 80/5/15 split suggests that the model had already learned most of the discriminative patterns by the 60% split.

4.4 XGBoost + PCA: XGBoost + PCA represents a hybrid machine learning approach that combines dimensionality reduction with ensemble learning. In many image classification problems, raw image data contains thousands of features, many of which may be redundant or highly correlated. Principal Component Analysis (PCA) is used to transform the original feature space into a smaller set of principal components that retain most of the important information. These reduced features are then used as input to the XGBoost classifier. By combining the strengths of PCA and XGBoost, this hybrid approach aims to improve computational efficiency while maintaining strong classification performance.

For the Brain MRI dataset, the feature space was reduced to **200** principal components, while for the Chest X-Ray dataset, **500** principal components were retained. These reduced feature representations were then used to train an XGBoost classifier. By focusing on the most informative features and eliminating unnecessary dimensions, the PCA + XGBoost pipeline aimed to improve computational efficiency while maintaining strong classification performance.

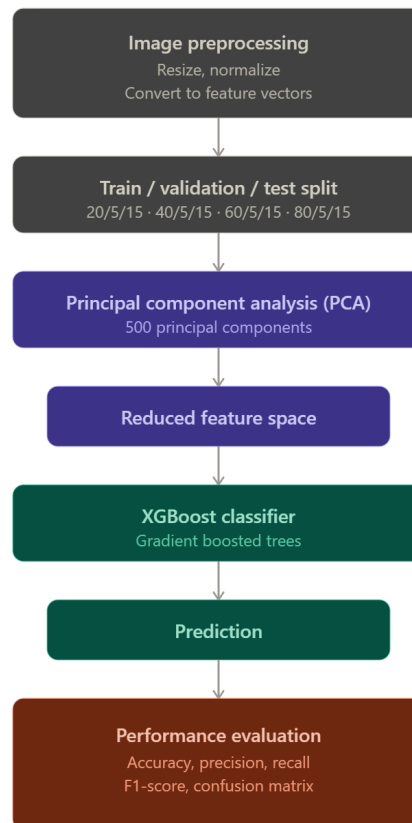


Figure 16: XGBoost + PCA pipeline flowchart

4.4.1 RESULTS

brain	Train/val/test split	Validation accuracy	Test accuracy	Macro precision	Macro recall	Macro F1-score	Best boosting round	Validation log loss
	20/5/15 (574/145/431)	70.34%	71.69%	0.74	0.67	0.67	67	0.800816
	40/5/15 (1146/145/431)	74.48%	79.12%	0.81	0.75	0.77	193	0.557101
	60/5/15 (1179/145/431)	83.45%	83.53%	0.86	0.79	0.81	199	0.374399
	80/5/15 (2294/145/431)	86.21%	84.69%	0.86	0.82	0.83	190	0.339677
chest	Train/val/test split	Validation accuracy	Test accuracy	Macro precision	Macro recall	Macro F1-score	Best boosting round	Validation log loss
	20/5/15 (1171/293/878)	94.20%	91.91%	0.92	0.87	0.89	179	0.175808
	40/5/15 (2342/293/878)	95.56%	94.42%	0.94	0.91	0.93	198	0.115352
	60/5/15 (3513/293/878)	95.56%	93.51%	0.93	0.90	0.91	185	0.120161
	80/5/15 (4684/293/878)	95.90%	93.96%	0.93	0.91	0.92	199	0.108535

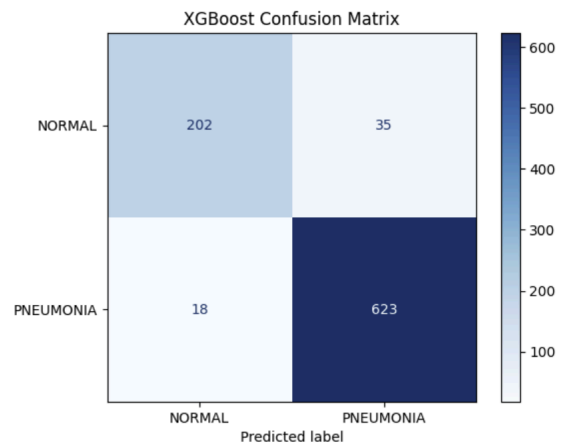
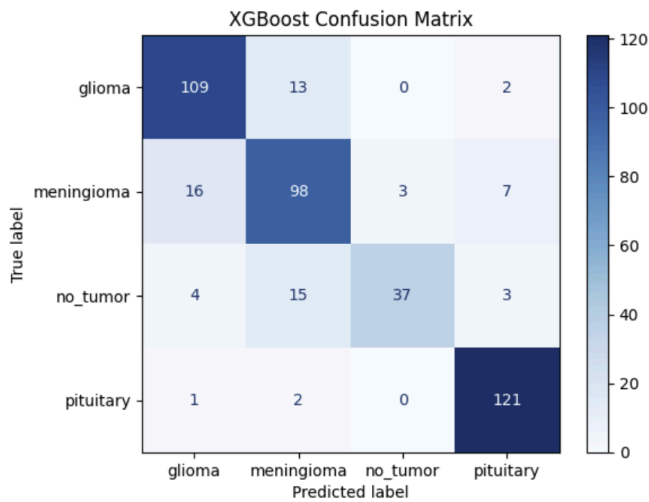


Figure 17: Confusion matrix of Brain MRI and Chest X-Ray (80/5/15 split)

4.4.2 INFERENCE

1. Validation log loss decreased consistently for both datasets as the training data increased, suggesting that the PCA-transformed features enabled the model to make more confident and reliable predictions.
2. The Chest X-Ray dataset maintained high performance across all training splits, achieving over **91%** test accuracy even with only **20%** training data. This indicates that the PCA-reduced features preserved most of the discriminative information present in the original images.
3. The gap between Brain MRI and Chest X-Ray performance was more pronounced, suggesting that dimensionality reduction had a greater impact on the more complex Brain MRI classification task than on the binary Chest X-Ray classification task.

4.5 XGBoost + VGG16: XGBoost + VGG16 is a hybrid machine learning approach that combines deep feature extraction with ensemble learning. VGG16 is a deep Convolutional Neural Network (CNN) pre-trained on the ImageNet dataset and is capable of learning rich hierarchical image features such as edges, textures, shapes, and complex visual patterns. Instead of using VGG16 for end-to-end classification, the network is used as a feature extractor, while XGBoost performs the final classification task. This approach leverages the powerful feature learning capability of deep learning models together with the robustness and efficiency of gradient-boosted decision trees.

VGG16 was used to extract high-level feature representations from both the Brain MRI and Chest X-Ray images. The extracted deep features were then provided as input to the XGBoost classifier, which learned to distinguish between the different image classes. By replacing manually engineered features with features learned by a pre-trained CNN, the model was able to capture more informative visual patterns present in the medical images. This hybrid approach aimed to combine the representational power of deep learning with the classification strength of XGBoost for medical image analysis.

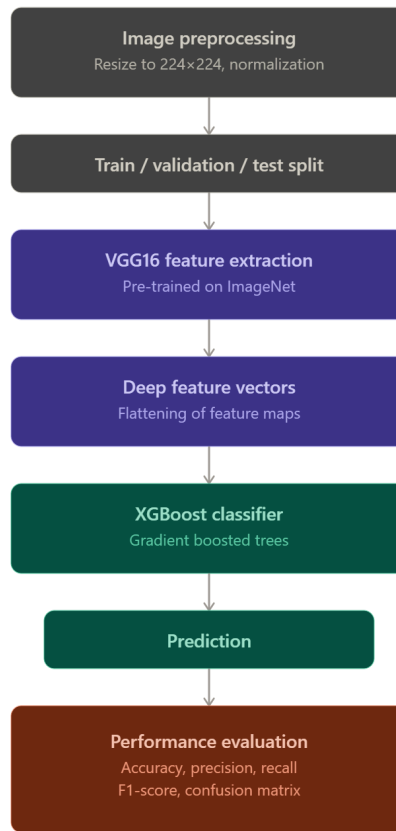


Figure 18: XGBoost + VGG16 pipeline flowchart

4.5.1 RESULTS

brain	Train/val/test split	Validation accuracy	Test accuracy	Macro precision	Macro recall	Macro F1-score	Best boosting round	Validation log loss
	20/5/15 (574/145/431)	86.21%	87.47%	0.87	0.88	0.88	197	0.361304
	40/5/15 (1146/145/431)	88.28%	91.18%	0.91	0.91	0.91	176	0.311531
	60/5/15 (1179/145/431)	88.97%	92.58%	0.93	0.92	0.93	174	0.270454
	80/5/15 (2294/145/431)	88.28%	93.27%	0.93	0.93	0.93	198	0.21112
chest	Train/val/test split	Validation accuracy	Test accuracy	Macro precision	Macro recall	Macro F1-score	Best boosting round	Validation log loss
	20/5/15 (1171/293/878)	94.20%	95.44%	0.95	0.94	0.94	71	0.13208
	40/5/15 (2342/293/878)	95.22%	95.90%	0.95	0.95	0.95	128	0.09907
	60/5/15 (3513/293/878)	96.25%	96.47%	0.96	0.95	0.96	97	0.082803
	80/5/15 (4684/293/878)	96.59%	96.70%	0.96	0.96	0.96	129	0.092618

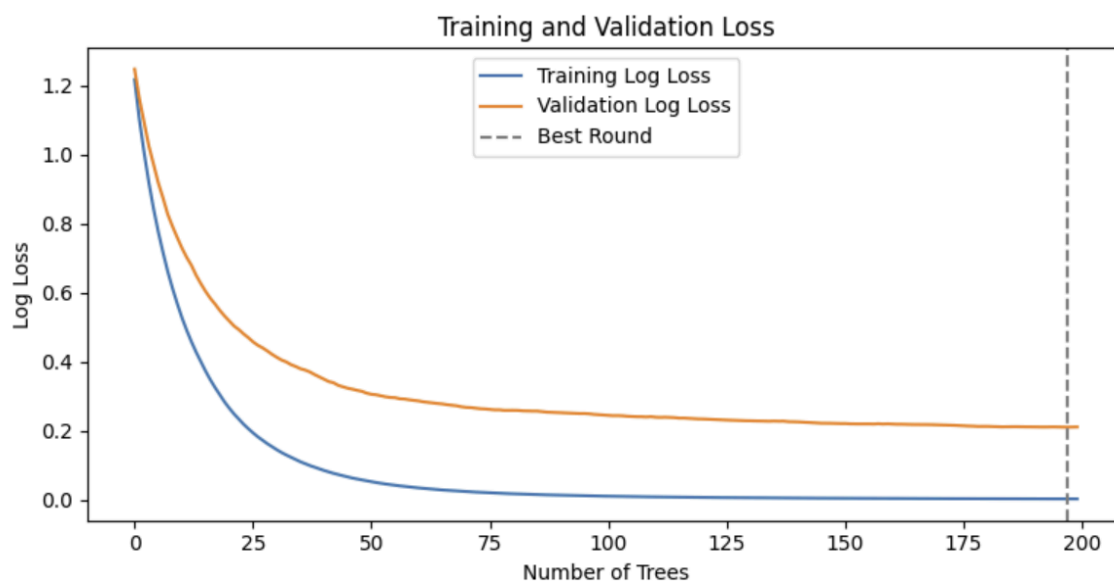
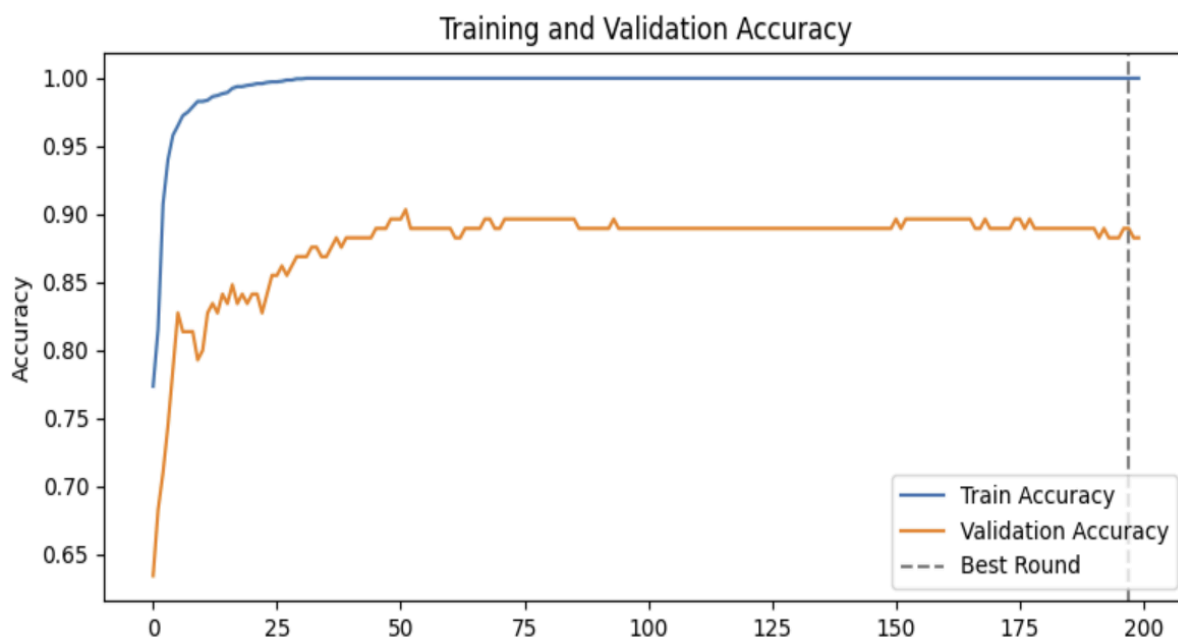


Figure 19 : validation accuracy curve and validation loss curve for brain MRI

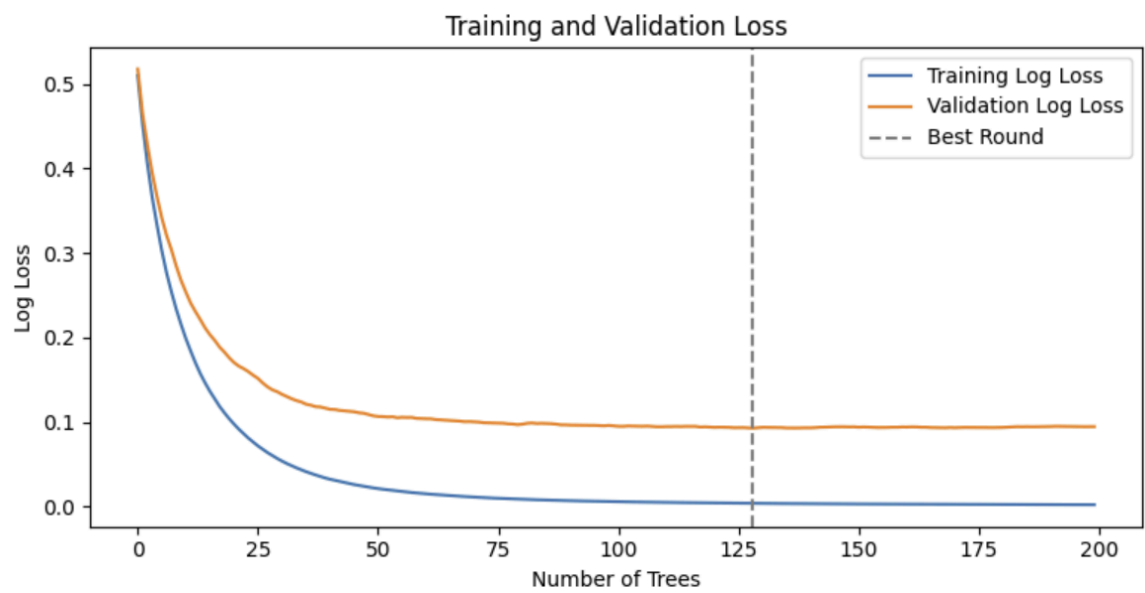
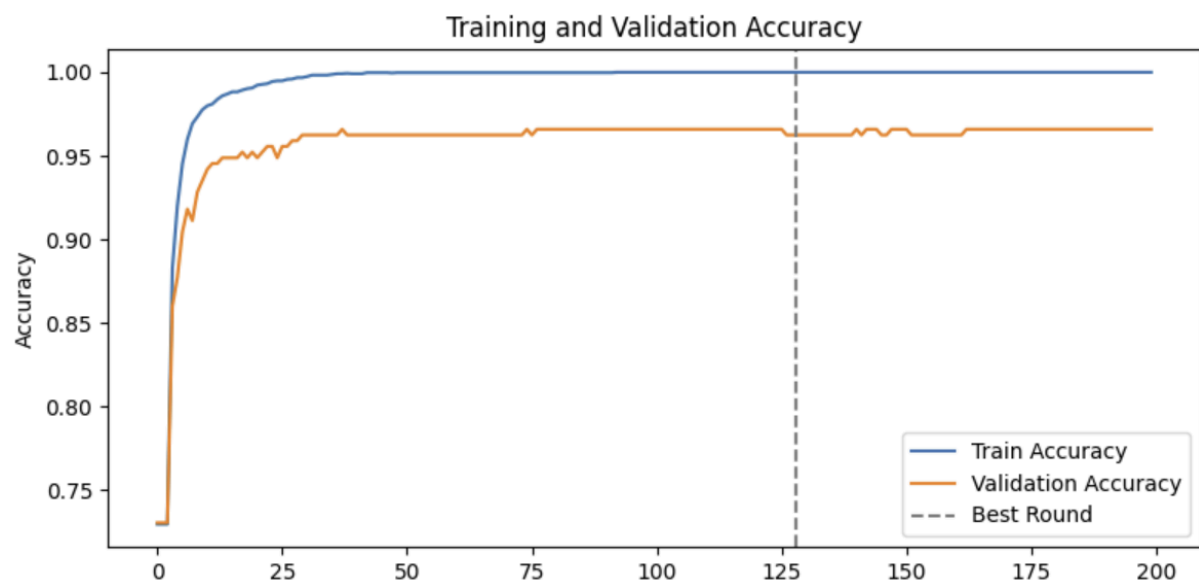


Figure 20 : Validation accuracy curve and validation loss curve for chest X-Ray

4.5.2 INFERENCE

1. Validation log loss generally decreased with increasing training data, suggesting that the VGG16 feature representations enabled XGBoost to make more confident and accurate predictions.
2. The Chest X-Ray dataset achieved its highest test accuracy of 96.70% at the 80/5/15 split, demonstrating that the deep features learned by VGG16 were highly effective in capturing disease-specific patterns present in X-Ray images.
3. Both Brain MRI and Chest X-Ray datasets showed a consistent improvement in test accuracy as the training data increased, indicating that the deep features extracted by VGG16 became more informative when larger training datasets were available.

5. OBSERVATIONS

1. The performance gap between SVM and CNN increased as the amount of training data increased, demonstrating CNN's ability to utilize larger datasets more effectively.
2. For the Brain MRI dataset, **XGBoost** achieved the highest test accuracy (**93.97%**), followed closely by **VGG16 + XGBoost** (**93.27%**), while **PCA + XGBoost** achieved the lowest accuracy (**84.69%**).
3. For the Chest X-Ray dataset, **VGG16 + XGBoost** achieved the highest test accuracy (**96.70%**), followed by **XGBoost** (**94.99%**) and **PCA + XGBoost** (**94.42%**).
4. PCA + XGBoost consistently produced the lowest performance among the three approaches, indicating a loss of discriminative information during dimensionality reduction.

5. VGG16 + XGBoost achieved the highest Precision, Recall, and F1-Score values on the Chest X-Ray dataset, making it the strongest overall performer.
6. XGBoost and VGG16 + XGBoost produced comparable results on the Brain MRI dataset, whereas VGG16 + XGBoost showed a clearer advantage on the Chest X-Ray dataset.
7. Overall, **VGG16 + XGBoost** emerged as the **best-performing hybrid approach**.

6. CONCLUSION

The experimental results demonstrated that model performance generally improved as the amount of training data increased. SVM + PCA achieved satisfactory classification performance but was consistently outperformed by the more advanced approaches. XGBoost proved to be the strongest traditional machine learning method, delivering high accuracy and robust performance across both datasets. Although PCA reduced the dimensionality of the feature space, it resulted in a decline in classification performance, particularly for the Brain MRI dataset.

CNN-based transfer learning using MobileNet achieved strong results and exhibited good generalization across different train-test splits. Among all evaluated approaches, the VGG16 + XGBoost hybrid model produced the most consistent and balanced performance, combining the feature extraction capability of deep learning with the classification strength of ensemble learning. Overall, the results indicate that approaches utilizing deep learned features are more effective than traditional feature-based methods, making hybrid and deep learning models the most suitable choices for the classification tasks considered in this project.